COURSE SLIDES

Certified Lean Six Sigma Yellow Belt (CLSSYB) Training

Course Code: C524

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer

?

Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Certified Lean Six Sigma practitioner — process improvement, quality and data-driven problem solving.

DELIVERS

Professional short courses on Lean Six Sigma, quality management, data and cloud.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

1. Your name and organisation / role.
2. A process in your work that frustrates you or your customers.
3. What you want to be able to improve after this course.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

Download Course Material

1. Sign in at the LMS-TMS portal
2. Open this course from your dashboard
3. Go to the Courseware tab
4. Download the Slides (PPT/PDF), Learner Guide and Lesson Plan
5. Keep them open alongside the labs as you work

Portal: https://lms-tms.tertiaryinfotech.com

SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

Day 1

- Day 1 — Six Sigma foundations, Define phase and process mapping
  - Topic 1: Six Sigma Foundations (Lab 1)
  - Topic 2: Define — Scope the Problem (Labs 2–5, 11)

Day 2

- Day 2 — Measure, Analyse, Improve and Control
  - Topic 3: Measure — Quantify Performance (Labs 6, 12)
  - Topic 4: Analyze — Find the Root Cause (Labs 7–8)
  - Topic 5: Improve — Fix the Cause (Labs 9, 13)
  - Topic 6: Control — Hold the Gain (Labs 10, 14)
- Daily timing
  - 9:30am–6:30pm · 1-hour lunch · tea breaks within training time

WHAT YOU'LL ACHIEVE

Learning Outcomes

1 **Define & Scope**
Define a project and establish the scope of work using Six Sigma's Define, Measure and Analyse phases.

2 **Lean & Six Sigma Concepts**
Apply Lean and Six Sigma concepts — value, waste, defects and variation — to a work process.

3 **Process Mapping**
Map a process using SIPOC, process maps and value stream maps to expose handoffs and waste.

4 **Data & Analysis**
Collect and analyse process data using check sheets, Pareto charts, run charts and basic metrics.

5 **Root Cause Analysis**
Identify root causes using 5 Whys, Fishbone analysis and evidence-based prioritisation.

6 **Improve & Control**
Recommend improvement and control actions to sustain gains, and prepare for certification.

LEARNING APPROACH

How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

CORE CONCEPTS

Lean Six Sigma Fundamentals

OVERVIEW

What is Lean Six Sigma?

1 **Lean removes waste**
Maximise customer value by eliminating the effort the customer would never pay for.

2 **Six Sigma reduces variation**
A data-driven method targeting 3.4 defects per million opportunities.

3 **Together: fast and accurate**
Lean Six Sigma delivers faster flow AND fewer defects, driven by evidence.

4 **Everyone has a belt role**
White, Yellow, Green, Black and Master Black Belt — the Yellow Belt supports.

THE BUILDING BLOCKS

The DMAIC Roadmap

1

Define

Scope the problem — VOC, CTQ, charter, problem statement, SIPOC.

2

Measure

Quantify today's performance — data collection, check sheets, yield and DPMO.

3

Analyze

Find the root cause — Pareto, run charts, 5 Whys and Fishbone.

4

Improve

Fix the cause — countermeasures, 5S, poka-yoke, standard work and piloting.

5

Control

Hold the gain — control plan, SOPs, visual management, A3 and handover.

THE LEAN SIX SIGMA MINDSET

Improve the process, not the people.

Every tool in this course points at the process: measure what it actually does, find why it does it, and change the system so the improvement holds.

COURSE ROADMAP

One Scenario, One Complete Package

Define & Measure

- Contoso Service Desk
- VOC → CTQ, SIPOC
- Data collection plan

Analyze

- Pareto and run charts
- 5 Whys and Fishbone
- Evidence-tested root cause

Improve & Control

- Countermeasures and 5S
- Control plan and SOPs
- A3 summary and handover

THE LEARNING ARC

How Every Lab Progresses

1. Every lab uses one continuous scenario — the Contoso Service Desk, where IT tickets are slow to be assigned.
2. Each lab produces a real artifact: a table, a map, a chart or a plan.
3. Each lab ends with a 'Check your work' step so you can verify your own output.
4. By the end your outputs form one complete improvement package, ready to reuse at work.

01

TOPIC 01

Six Sigma Foundations

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

COURSE COVERAGE 15%

Key Concepts — Six Sigma Foundations

1

What is Quality

Quality is conformance to customer requirements — not merely defect-free output.

2

What is Lean

A method to maximise customer value by systematically removing waste and improving flow.

3

What is Six Sigma

A data-driven method to reduce variation and defects — targeting 3.4 defects per million.

4

Lean Six Sigma

The combined method: faster flow AND fewer defects, driven by data.

5

Belt roles

White, Yellow, Green, Black and Master Black Belt — who does what on a project.

6

The DMAIC roadmap

Define, Measure, Analyze, Improve, Control — the disciplined improvement path.

WHAT YOU'LL DO

Hands-On Labs — Six Sigma Foundations

Lab 1

- Yellow Belt Role, Certification Paths, and Improvement Scenario

TOPIC 01 · HANDS-ON

Yellow Belt Role, Certification Paths, and Improvement Scenario

LAB 1

Establish what a Lean Six Sigma Yellow Belt is accountable for, how the belt pathway progresses from White to Master Black Belt, and choose a small, manageable improvement scenario to carry through the whole course.

You'll build

A Yellow Belt responsibility table and a selected improvement scenario.

Tools: Belt role matrix, CSSC certification pathway, project selection criteria

LAB 1

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 1 OF 5

1

Define the Yellow Belt role — list at least four responsibilities and your contribution to each.

LAB 1

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 2 OF 5

2

Compare the White, Yellow, Green, Black and Master Black Belt pathways in a table.

LAB 1

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 3 OF 5

3

Review the Contoso Service Desk scenario: tickets are slow to be assigned and status is unclear.

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 4 OF 5

4

Test your scenario against the 'good project' criteria — day-to-day work, manageable, aligned to business goals, data available.

LAB 1

Yellow Belt Role, Certification Paths, and Improvement Scenario

STEP 5 OF 5

5

Record why a Yellow Belt supports rather than leads this improvement.

Yellow Belt Role, Certification Paths, and Improvement Scenario

✅ Test it

You can state the Yellow Belt role in one sentence and justify your scenario against all four selection criteria.

TOPIC RECAP

Recap — Six Sigma Foundations

1. You can now: Explain the Yellow Belt role and select a suitable improvement scenario (A1)

02

TOPIC 02

Define — Scope the Problem

VOC · CTQ · Project charter · Problem statement · Scope · SIPOC · Process mapping

COURSE COVERAGE 25%

Key Concepts — Define — Scope the Problem

1

Voice of the Customer

Capture customer needs in their own words before deciding anything.

2

Critical to Quality

Translate each VOC need into a specific, measurable CTQ requirement.

3

Project charter

Problem statement, goal, scope, team and benefit — the project's contract.

4

Problem statement

Process, time period, measurable issue and impact — and never a solution.

5

SIPOC

The macro as-is map: Suppliers, Inputs, Process, Outputs, Customers.

6

Process mapping

Flowcharts and swimlanes expose the handoffs where delay and defects are born.

WHAT YOU'LL DO

Hands-On Labs — Define — Scope the Problem

Labs 2–3

- Lean, Six Sigma, Waste, Voice of Customer, and Value
- SIPOC, Process Mapping, Handoffs, and SME Support

Labs 4–5

- PDCA Small Improvement Project Charter
- DMAIC Overview, Problem Statement, Scope, and Stakeholders

Lab 11

- Affinity Diagram and Kano Analysis

Lean, Six Sigma, Waste, Voice of Customer, and Value

LAB 2

Separate what the customer actually values from the waste that surrounds it. Capture the Voice of the Customer, translate it into measurable CTQ requirements, and run a waste walk using the eight wastes (DOWNTIME).

You'll build

A VOC-to-CTQ translation table, a value-added analysis, and a waste walk log.

Tools: VOC, CTQ tree, DOWNTIME eight wastes, value-added analysis

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 1 OF 6

1

Capture the Voice of the Customer — record at least five verbatim customer statements.

LAB 2

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 2 OF 6

2

Translate each VOC statement into a need, then into a measurable CTQ requirement.

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 3 OF 6

3

Classify each process activity as value-added, business-value-added or non-value-added.

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 4 OF 6

4

Conduct a waste walk and tag each observation against the eight wastes (DOWNTIME).

LAB 2

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 5 OF 6

5

Distinguish a defect (output fails CTQ) from waste (effort the customer will not pay for).

LAB 2

Lean, Six Sigma, Waste, Voice of Customer, and Value

STEP 6 OF 6

6

Identify which single waste type appears most often in your scenario.

Lean, Six Sigma, Waste, Voice of Customer, and Value

✅ Test it

Every CTQ is measurable with a target, and each waste observation is tagged to one of the eight waste types.

TOPIC 02 · HANDS-ON

SIPOC, Process Mapping, Handoffs, and SME Support

LAB 3

Build the macro 'as-is' view with SIPOC, then drill into a detailed process map that shows every actor, system and handoff — the points where delay and defects are actually created.

You'll build

A completed SIPOC and a detailed process map with pain points marked.

Tools: SIPOC, process flowchart, swimlane map, standard process symbols

LAB 3

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 1 OF 6

1

Set the process boundaries — agree the explicit start and stop points first.

LAB 3

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 2 OF 6

2

Build the SIPOC: Suppliers, Inputs, Process (5-7 high-level steps), Outputs, Customers.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 3 OF 6

3

Expand into a detailed process map: Step, Actor, Activity, System, Handoff.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 4 OF 6

4

Draw the same flow as a swimlane map so each actor's lane makes handoffs obvious.

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 5 OF 6

5

Mark pain points — delays, rework loops, unclear ownership and handoff risks.

LAB 3

SIPOC, Process Mapping, Handoffs, and SME Support

STEP 6 OF 6

6

Prepare SME notes for the Green Belt: what you observed and what needs validation.

LAB 3 · VERIFY

SIPOC, Process Mapping, Handoffs, and SME Support

✅ Test it

Your SIPOC has all five columns populated and every handoff on the detailed map has a named owner on both sides.

PDCA Small Improvement Project Charter

LAB 4

Not every improvement needs a full DMAIC project. Use PDCA to charter a small, fast improvement that a Yellow Belt can run under guidance — with a problem statement, a measurable goal and a defined check point.

You'll build

A one-page PDCA charter with problem statement, goal, scope and success measure.

Tools: PDCA cycle, problem/goal statements, scope in-out, stakeholder list

LAB 4

PDCA Small Improvement Project Charter

STEP 1 OF 6

1

Map the four PDCA phases to concrete actions for your scenario.

LAB 4

PDCA Small Improvement Project Charter

STEP 2 OF 6

2

Write the problem statement: process, time period, measurable issue and impact — no solution.

LAB 4

PDCA Small Improvement Project Charter

STEP 3 OF 6

3

Write a measurable goal statement: metric, baseline, target and date.

LAB 4

PDCA Small Improvement Project Charter

STEP 4 OF 6

4

Define scope with an explicit in-scope / out-of-scope table to prevent scope creep.

LAB 4

PDCA Small Improvement Project Charter

STEP 5 OF 6

5

Identify stakeholders and the decision you will make after the Check phase.

LAB 4

PDCA Small Improvement Project Charter

STEP 6 OF 6

6

Confirm the improvement is small enough to test within two weeks.

LAB 4 · VERIFY

PDCA Small Improvement Project Charter

✅ Test it

Your goal statement contains a metric, a baseline, a target and a date, and your problem statement names no solution.

TOPIC 02 · HANDS-ON

DMAIC Overview, Problem Statement, Scope, and Stakeholders

LAB 5

Step up from PDCA to the full DMAIC roadmap. Clarify what each phase delivers, where a Yellow Belt contributes most, and prepare defensible Define-phase information for the project leader.

You'll build

A DMAIC phase table, refined problem statement, stakeholder map and benefit estimate.

Tools: DMAIC roadmap, project charter, stakeholder analysis, CTQ linkage

LAB 5

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 1 OF 6

1

Summarise all five DMAIC phases: purpose, key deliverable and Yellow Belt support role.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 2 OF 6

2

Refine the problem statement so it is specific, measurable and solution-free.

LAB 5

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 3 OF 6

3

Identify stakeholders and classify each by influence and interest.

LAB 5

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 4 OF 6

4

Define the project scope and state the expected benefit in business terms.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 5 OF 6

5

Link the problem back to the CTQ requirements captured in Lab 2.

LAB 5

DMAIC Overview, Problem Statement, Scope, and Stakeholders

STEP 6 OF 6

6

Confirm which DMAIC phases a Yellow Belt can support most strongly.

DMAIC Overview, Problem Statement, Scope, and Stakeholders

✅ Test it

Your problem statement passes the 'no solution named' test and every stakeholder has a defined engagement approach.

TOPIC 02 · HANDS-ON

Affinity Diagram and Kano Analysis

LAB 11

Two Define-phase tools from the original course. Use an Affinity Diagram to cluster unstructured VOC input into themes, then apply Kano Analysis to classify requirements as Must-Be, One-Dimensional or Delighter.

You'll build

An Affinity Diagram of clustered VOC themes and a Kano classification table.

Tools: Affinity Diagram, Kano Analysis, VOC clustering

LAB 11

Affinity Diagram and Kano Analysis

STEP 1 OF 4

1

Write each VOC statement on a separate note — one idea per note.

LAB 11

Affinity Diagram and Kano Analysis

STEP 2 OF 4

2

Cluster related notes into natural groups without talking, then name each group.

LAB 11

Affinity Diagram and Kano Analysis

STEP 3 OF 4

3

Classify each requirement as Must-Be, One-Dimensional or Delighter.

LAB 11

Affinity Diagram and Kano Analysis

STEP 4 OF 4

4

Plot the requirements on the Kano diagram and identify where to invest first.

Affinity Diagram and Kano Analysis

✅ Test it

Every VOC note sits in exactly one named affinity group and carries a Kano classification.

TOPIC RECAP

Recap — Define — Scope the Problem

1. You can now: Apply Lean and Six Sigma concepts — value, waste, defects, variation (K1, A2)
2. You can now: Map a process with SIPOC and a detailed process map to expose handoffs (A2, A3)
3. You can now: Charter a small improvement using PDCA with a measurable goal (A1, A2)
4. You can now: Support the Define phase of a DMAIC project (A1, A2)
5. You can now: Organise customer requirements and classify them by satisfaction impact (K2)

03

TOPIC 03

Measure — Quantify Performance

The 8 wastes · Data types · Data collection plan · Check sheets · Yield · DPMO · Sigma level

COURSE COVERAGE 25%

Key Concepts — Measure — Quantify Performance

1

The 8 wastes — DOWNTIME

Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.

2

Types of data

Continuous vs discrete; nominal vs ordinal — the data type drives the tool choice.

3

Data collection plan

What, who, when and how — with operational definitions to remove ambiguity.

4

Check sheets

A simple structured form is the most reliable way to capture process data.

5

Process metrics

Yield, DPU, DPO and DPMO quantify how well the process actually performs.

6

Sigma level

Convert DPMO into a sigma level to benchmark the process against Six Sigma.

WHAT YOU'LL DO

Hands-On Labs — Measure — Quantify Performance

Lab 6

- Data Collection, KPIs, Check Sheets, and Basic Metrics

Lab 12

- Value Stream Map and Takt Time

TOPIC 03 · HANDS-ON

Data Collection, KPIs, Check Sheets, and Basic Metrics

LAB 6

Replace anecdote with evidence. Define KPIs with unambiguous operational definitions, design a check sheet, plan the sampling approach, and understand which data type you are collecting — because the data type drives the tool.

You'll build

A data collection plan, an operational definition set and a working check sheet.

Tools: KPI definition, operational definitions, check sheets, sampling, data types

LAB 6

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 1 OF 6

1

Classify your data: continuous vs discrete, nominal vs ordinal — and why it matters.

LAB 6

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 2 OF 6

2

Define KPIs with operational definitions, data source and collection frequency.

LAB 6

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 3 OF 6

3

Design a check sheet capturing date, ticket ID, type, assignment time and defect category.

LAB 6

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 4 OF 6

4

Define mutually exclusive defect categories so every observation lands in exactly one.

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 5 OF 6

5

Plan sampling: how many, how often, by whom — and identify possible bias.

LAB 6

Data Collection, KPIs, Check Sheets, and Basic Metrics

STEP 6 OF 6

6

State which KPI best reflects the customer pain point from your VOC work.

Data Collection, KPIs, Check Sheets, and Basic Metrics

✅ Test it

Two different people reading your operational definition would record the same value for the same event.

TOPIC 03 · HANDS-ON

Value Stream Map and Takt Time

LAB 12

Extend process mapping into Lean metrics. Map the value stream with process and wait times, calculate process cycle efficiency, then compute takt time to see whether the process can meet customer demand.

You'll build

A value stream map with lead time, process time and a calculated takt time.

Tools: Value stream mapping, lead time, WIP, takt time, cycle efficiency

LAB 12

Value Stream Map and Takt Time

STEP 1 OF 4

1

Map the value stream: each step with its process time and the wait time between steps.

LAB 12

Value Stream Map and Takt Time

STEP 2 OF 4

2

Total the value-added time and the lead time, then compute process cycle efficiency.

LAB 12

Value Stream Map and Takt Time

STEP 3 OF 4

3

Calculate takt time = available working time / customer demand.

LAB 12

Value Stream Map and Takt Time

STEP 4 OF 4

4

Compare cycle time against takt time to identify the bottleneck step.

Value Stream Map and Takt Time

✅ Test it

Your lead time equals the sum of all process and wait times, and takt time is expressed per unit.

TOPIC RECAP

Recap — Measure — Quantify Performance

1. You can now: Plan and execute data collection against defined quality standards (A4)

2. You can now: Quantify flow, lead time and takt time across the value stream (A3, A4)

04

TOPIC 04

Analyze — Find the Root Cause

Variation · Pareto · Run charts · 5 Whys · Fishbone · Multi-voting · Evidence

COURSE COVERAGE 20%

Key Concepts — Analyze — Find the Root Cause

1

Variation

Common cause is built into the process; special cause is an external, assignable signal.

2

Pareto analysis

The 80/20 rule separates the vital few causes from the trivial many.

3

Run charts

Plot the metric over time to reveal trend, shift, cluster and oscillation.

4

5 Whys

Ask why repeatedly to drill from the symptom down to an actionable cause.

5

Fishbone diagram

Organise candidate causes by category — Manpower, Method, Machine, Material, Measurement.

6

Evidence testing

A cause is only a root cause when the data you collected supports it.

WHAT YOU'LL DO

Hands-On Labs — Analyze — Find the Root Cause

Lab 7

- Pareto, Run Charts, Variation, Yield, DPU, and DPMO

Lab 8

- Root Cause Analysis with 5 Whys, Fishbone, and Evidence

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

LAB 7

Turn collected data into decisions. Apply the Pareto principle to find the vital few categories, use a run chart to see behaviour over time, and calculate the standard Six Sigma process metrics.

You'll build

A Pareto chart, a run chart and calculated yield, DPU, DPO, DPMO and sigma level.

Tools: Pareto Chart tool (collaborative

LAB 7

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 1 OF 8

1

Build the Pareto table: category, count, percentage and cumulative percentage.

LAB 7

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 2 OF 8

2

In your group, open the collaborative Pareto tool — one member creates the session and shares the code, then everyone joins, brainstorms and votes to produce the live Pareto chart.

LAB 7

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 3 OF 8

3

Identify the vital few categories driving about 80% of the problem.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 4 OF 8

4

Export your assignment-time data as CSV and plot a run chart in NovaSPC.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 5 OF 8

5

Interpret the run chart for trend, shift, cluster and oscillation patterns.

LAB 7

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 6 OF 8

6

Calculate yield, DPU, DPO and DPMO from your defect data.

LAB 7

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 7 OF 8

7

Convert DPMO to a sigma level and interpret what it says about the process.

LAB 7

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

STEP 8 OF 8

8

Distinguish common-cause from special-cause variation and why the response differs.

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

✅ Test it

Your cumulative percentage column reaches 100%, and you can state the sigma level with the DPMO it came from.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

LAB 8

Move from symptom to cause. Apply the 5 Whys to drill past the obvious, organise candidate causes on a Fishbone diagram, then test each candidate against the evidence you actually collected.

You'll build

A completed 5 Whys chain, a Fishbone diagram and an evidence-tested cause shortlist.

Tools: 5 Whys tool, Fishbone tool, 5M categories, brainstorming, multi-voting

LAB 8

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 1 OF 7

1

Write the problem statement precisely — the effect you are explaining.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 2 OF 7

2

Complete a 5 Whys chain with the online 5 Whys tool, continuing until you reach an actionable cause.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 3 OF 7

3

Build a Fishbone diagram with the online Fishbone tool using the 5M categories: Manpower, Method, Machine, Material, Measurement.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

4

Brainstorm candidate causes into each category — no evaluation during generation.

LAB 8

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 5 OF 7

5

Use multi-voting to shortlist the most likely causes as a team.

LAB 8

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 6 OF 7

6

Test each shortlisted cause against your Lab 7 data — does the evidence support it?

LAB 8

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

STEP 7 OF 7

7

State why the team must not jump straight to solutions.

Root Cause Analysis with 5 Whys, Fishbone, and Evidence

✅ Test it

Each shortlisted root cause is supported by named evidence, and your 5 Whys chain ends at something you can act on.

TOPIC RECAP

Recap — Analyze — Find the Root Cause

1. You can now: Analyse process performance data to quantify and prioritise (A3, A4)

2. You can now: Identify root causes of variation using structured analysis (A3)

05

TOPIC 05

Improve — Fix the Cause

Countermeasures · 5S · Poka-Yoke · Standard work · Kaizen · Solution selection · Piloting

COURSE COVERAGE 10%

Key Concepts — Improve — Fix the Cause

1

Generating solutions

Brainstorm widely against the proven root cause before evaluating anything.

2

Solution selection

Score candidate solutions against weighted criteria before committing.

3

5S

Sort, Set in order, Shine, Standardise, Sustain — for physical and digital work.

4

Poka-Yoke

Mistake proofing makes the error difficult or impossible to make in the first place.

5

Standard work

Document the improved method so anyone can repeat it the same way.

6

Piloting

Test the change at small scale to expose issues before full rollout.

WHAT YOU'LL DO

Hands-On Labs — Improve — Fix the Cause

Lab 9

- Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

Lab 13

- Solution Selection Matrix, Benchmarking, and FMEA

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

LAB 9

Generate and select countermeasures that attack the root cause you proved — not the symptom. Apply the core Lean improvement tools: 5S, poka-yoke, visual management, standard work and Kaizen.

You'll build

A prioritised countermeasure set with a 5S plan, a poka-yoke design and standard work.

Tools: Brainstorming, 5S, poka-yoke, visual management, standard work, Kaizen

LAB 9

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 1 OF 7

1

Generate countermeasures for each proven root cause — quantity before quality.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 2 OF 7

2

Prioritise using an impact vs effort matrix to find the quick wins.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 3 OF 7

3

Apply 5S thinking — Sort, Set in order, Shine, Standardise, Sustain — including to digital work.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 4 OF 7

4

Design one poka-yoke that makes the error difficult or impossible to make.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

5

Draft standard work so the improved method is repeatable by anyone.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 6 OF 7

6

Plan a Kaizen event or pilot to test the countermeasure at small scale first.

LAB 9

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

STEP 7 OF 7

7

Distinguish a containment countermeasure from a permanent solution.

Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

✅ Test it

Every countermeasure traces back to a proven root cause, and your poka-yoke prevents rather than detects the error.

Solution Selection Matrix, Benchmarking, and FMEA

LAB 13

Three Improve-phase tools from the original course. Score solutions against weighted criteria, benchmark against outstanding practice, and use FMEA to find how a proposed change could fail before it does.

You'll build

A scored solution selection matrix, a benchmarking summary and an FMEA with RPN.

Tools: Solution selection matrix, benchmarking, FMEA, RPN scoring

LAB 13

Solution Selection Matrix, Benchmarking, and FMEA

STEP 1 OF 5

1

Agree the selection criteria and weight each by importance.

Solution Selection Matrix, Benchmarking, and FMEA

STEP 2 OF 5

2

Score every candidate solution against the weighted criteria and rank the results.

Solution Selection Matrix, Benchmarking, and FMEA

STEP 3 OF 5

3

Benchmark your process against an internal or external reference and note the gap.

Solution Selection Matrix, Benchmarking, and FMEA

STEP 4 OF 5

4

Build an FMEA: failure mode, effect, cause, then score Severity, Occurrence and Detection.

LAB 13

Solution Selection Matrix, Benchmarking, and FMEA

STEP 5 OF 5

5

Calculate RPN = S x O x D and address the highest-RPN failure modes first.

Solution Selection Matrix, Benchmarking, and FMEA

✅ Test it

Your matrix ranks solutions by weighted score, and every FMEA row has an RPN and an action for the highest scores.

TOPIC RECAP

Recap — Improve — Fix the Cause

1. You can now: Recommend improvement actions that address the proven root cause (A5)

2. You can now: Evaluate and de-risk candidate solutions before implementation (A5, K2)

06

TOPIC 06

Control — Hold the Gain

Control plan · Visual management · SOPs · Huddles · A3 · Handover · Certification

COURSE COVERAGE 5%

Key Concepts — Control — Hold the Gain

1 **Control plan**
Metric, target, monitoring method, frequency, owner and reaction plan.

2 **Process control**
A process is in control when it varies consistently within expected limits.

3 **Visual management**
Boards and dashboards keep the improved performance visible to everyone.

4 **Standard operating procedures**
Written instructions that lock in the improved method.

5 **Team huddles**
Short, regular stand-ups that surface problems while they are still small.

6 **A3 and handover**
One-page storytelling to summarise, hand over and sustain the improvement.

WHAT YOU'LL DO

Hands-On Labs — Control — Hold the Gain

Lab 10

- Control Plan, A3 Summary, Handover, and Certification Readiness

Lab 14

- Descriptive Statistics and Implementation Planning

Control Plan, A3 Summary, Handover, and Certification Readiness

LAB 10

Hold the gain. Build a control plan that names the metric, the target, the monitoring frequency, the owner and the reaction plan — then tell the whole improvement story on a single A3 page and hand it over.

You'll build

A control plan, an A3 one-page summary, a handover checklist and a readiness plan.

Tools: Control plan, SPC thinking, visual management, A3 report, handover

LAB 10

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 1 OF 6

1

Create the control plan: metric, target, monitoring method, frequency, owner, reaction plan.

LAB 10

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 2 OF 6

2

Define the visual management approach — board, dashboard or huddle — that keeps it visible.

Control Plan, A3 Summary, Handover, and Certification Readiness

3

Specify the reaction plan: exactly what happens when a metric falls outside target.

LAB 10

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 4 OF 6

4

Write the A3 summary: background, current state, goal, analysis, countermeasures, results, follow-up.

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 5 OF 6

5

Prepare the handover checklist so the process owner can sustain it without you.

LAB 10

Control Plan, A3 Summary, Handover, and Certification Readiness

STEP 6 OF 6

6

Complete your personal certification readiness plan — which topics need most review.

Control Plan, A3 Summary, Handover, and Certification Readiness

✅ Test it

Every control plan row has a named owner and a reaction plan, and your A3 fits on one page.

TOPIC 06 · HANDS-ON

Descriptive Statistics and Implementation Planning

LAB 14

Close the loop. Compute the measures of central tendency and dispersion that describe your data, then convert the selected countermeasures into a dated implementation plan with owners and barriers identified.

You'll build

A descriptive statistics summary and a dated implementation plan.

Tools: Mean, median, range, standard deviation, implementation planning

LAB 14

Descriptive Statistics and Implementation Planning

STEP 1 OF 4

1

Compute the mean and median for your assignment-time data and compare them.

Descriptive Statistics and Implementation Planning

STEP 2 OF 4

2

Compute the range and standard deviation to describe the spread.

LAB 14

Descriptive Statistics and Implementation Planning

STEP 3 OF 4

3

Explain what the mean-versus-median difference reveals about outliers and skew.

Descriptive Statistics and Implementation Planning

STEP 4 OF 4

4

Write the implementation plan: action, owner, date, barriers and mitigation.

Descriptive Statistics and Implementation Planning

✅ Test it

You can explain why the mean and median differ in your data, and every implementation action has an owner and a date.

TOPIC RECAP

Recap — Control — Hold the Gain

1. You can now: Recommend control actions to sustain process performance (A5)

2. You can now: Summarise data numerically and plan the rollout (A4, A5)

WRAP-UP

Course Summary & Next Steps

LEARNING OUTCOMES

What You Achieved

1

Define & Scope

Define a project and establish the scope of work using Six Sigma's Define, Measure and Analyse phases.

2

Lean & Six Sigma Concepts

Apply Lean and Six Sigma concepts — value, waste, defects and variation — to a work process.

3

Process Mapping

Map a process using SIPOC, process maps and value stream maps to expose handoffs and waste.

4

Data & Analysis

Collect and analyse process data using check sheets, Pareto charts, run charts and basic metrics.

5

Root Cause Analysis

Identify root causes using 5 Whys, Fishbone analysis and evidence-based prioritisation.

6

Improve & Control

Recommend improvement and control actions to sustain gains, and prepare for certification.

NEXT STEPS

Continuing Your Lean Six Sigma Journey

1. Run a small improvement in your own workplace using the DMAIC roadmap and the templates from this course.

2. Progress towards Green Belt — deeper statistics, hypothesis testing and full project leadership.

3. Build the habit: measure before you change, and verify the gain held after you did.

4. Share the A3 one-pager with your team so the improvement is visible and sustained.

WRAP-UP

Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.

2. Apply the techniques to a real process or project in your own organisation.

3. Your course material stays available at https://lms-tms.tertiaryinfotech.com/.

4. Explore the follow-on courses at www.tertiarycourses.com.sg.

IMPROVE THE PROCESS

Thank You!

You can now scope, measure, analyse, improve and control a real process improvement using the DMAIC roadmap.